

Lenard Dome

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RESEARCH AREAS

Computational Modelling of the Complexity of Cognition and Behaviour

I developed a formal framework to benchmark formal model heterogeneity against human heterogeneity. The most novel aspect of this approach is to put a large emphasis on model predictions – events the models say will happen but we did not observe in the human data. This resulted in a new measure of model flexibility that allows us to see, quantify, and explore how unhuman models really are.

Categorization, Learning, Social Categorization

I carried out extensive research on irrational phenomena of categorization with a main focus on evaluating attentional theories of category learning. Taking an experimental psychologist's approach, I developed novel paradigms to test the validity, accuracy and consequences of those formal theories. I both confirmed and disconfirmed multiple model predictions as I increased the family of findings attentional theories have to explain in category learning.

Open Science and Open Software

I am a big promoter and supporter of Open Science practices. Throughout my scientific career, I hosted the ReproducibiliTea journal club for more than two years. I developed multiple R packages to nurture collaboration by implementing general-purpose methods and computational models with reproducible simulations. I delivered training and peer-to-peer support that helped peers and senior staff to adopt Open Science Practices. I also develop freely-available online training/teaching materials.

GRANTS AND AWARDS

Awards and Honors

- 2020 **Awarded** Experimental Psychological Society / British Science Association Undergraduate Project. Prize
[See the post on the EPS website.](#)
- 2019 **Awarded** Best Dissertation in Topics of Cognitive Psychology by School of Psychology, University of Plymouth.

Grants and Fellowships

- 2019 **Awarded** Doctoral Training Studentship (1+3) fully funded by the Economic and Social Research Council (ESRC) in collaboration with the South West Doctoral Training Partnership.

PUBLICATIONS

- in prep. **Dome, L., Wills, A.J.** “g-distance: A formal framework to compare model and human heterogeneity.”
- Dome, L., Inkster, A.B., Wills, A.J.** “It is taxing to be irrational: Clearing confounds from concurrent load and the inverse base-rate effect.”

- in review **Dome, L.**, Wills, A. J. “Errorless irrationality: removing error-driven components from the inverse base-rate effect paradigm.”
- 2021 **Dome, L.**, Edmunds, C. E. R., Wills, A. J. “SUSTAIN captures category learning, recognition, and hippocampal activation in a unidimensional vs information-integration task.” *Proceedings of the 43rd Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society. <https://escholarship.org/uc/item/5r98q3dr>
- Spicer, S. G., Wills, A. J., Jones, P. M., Mitchell, C., & **Dome, L.** “Representing uncertainty in the Rescorla-Wagner model: blocking, the redundancy effect, and outcome base rate.” *Open Journal of Experimental Psychology and Neuroscience*. doi.org/10.46221/ojepn.2021.6623
- 2019 Wimmer, M. C., **Dome, L.**, Wennekers, T. & Hancock, P. J. B. “Is the letter cancellation task a suitable index of ego-depletion? Empirical and conceptual issues.” *Social Psychology*. doi.org/10.1027/1864-9335/a000393

RESEARCH SOFTWARE

-  **psp** *maintainer, developer* Implements an n -dimensional parameter space partitioning algorithm for evaluating the global behaviour of formal computational models as described by Pitt, Kim, Navarro and Myung (2006). Currently has more than 5K downloads. [\[source code\]](#) [\[cran\]](#)
-  **catlearn** *senior developer* Catlearn is an archive of formal models of categorization and learning, plus benchmark datasets to test them against. It currently houses 14 formal mathematical models of learning and categorization. I have contributed 8 models to the package. Currently has more than 37K downloads. [\[source code\]](#) [\[cran\]](#)

INVITED TALKS

- 2021 Clearing confounds from the inverse base-rate effect: Irrationality and concurrent load. “2020 British Science Association / Experimental Psychological Society Undergraduate Project Prize Talk.” 2021. [\[video\]](#).
- The role of base rates in categorizing ambiguity and forming stereotypes. “Liverpool John Moores University: Cognitive and Affective Neuroscience Group Seminar.” 2021.

CONFERENCE PRESENTATIONS

- 2022 22nd Associative Learning Symposium: Gregynog Hall, Wales, UK.
- 2021 Cognitive Science Conference: Online.
- 2019 The British Psychological Society: South West Undergraduate Conference, Plymouth, UK.

TECHNICAL SKILLS

Linear and Non-linear Optimisation: *grid-search global optimisation, differential evolutionary algorithms for global optimisation, gradient descent on error.*

Global Model Behaviour, Data Discretization: *n -dimensional Parameter Space Partitioning.*

Computational Modelling, A.I.: *Embedding, Sequence Modelling, Feed-forward and Convolutional Neural Networks, Adaptive and Static Clustering Models.*

Data Science: *Bayesian Parameter Estimation and Model Comparison, Bayesian Linear Modelling, Supervised clustering, String Metrics (e.g. Levenshtein distance), Bayes Factor, Markov Chain Monte Carlo Sampling Methods, Parametric and Non-parametric Frequentist Methods, Big Data.*

Programming and Computing

R: *computational modelling, data analysis, programming, algorithms, statistical analysis*

Python: *experimental design, programming, computational modelling, statistical analysis*

C++: *computational modelling, statistical analysis, Rcpp and RcppArmadillo*

Javascript: *jsPsych for experimental design*

CSS: *webdesign*

UNIX: *Bash, CLI, Ubuntu Server, ssh*

High Performance Computing: *Slurm, Microsoft Azure, Google Cloud Computing*

ACADEMIC POSITIONS

2022 - 2023 Visiting Lecturer, Tutor and Specialist at the University of Plymouth.

Delivering and designing workshops on research methods, statistics and programming for peers, doctoral candidates and undergraduate students.

2019 - 2023 Teaching Assistant at University of Plymouth.

Assisting in the delivery of undergraduate modules on research methods with R.

2017 - 2019 Research Assistant/Apprentice. Line Manager: Andy Wills.

Collecting data in categorization experiments and programming implementations of formal computational models.

2017 - 2018 Research Assistant/Apprentice. Line Manager: Marina Wimmer.

Data collection and analysis.

EDUCATION

2020 - Summer 2023: Ph.D. Psychology. School of Psychology, Plymouth University.

The Studentship provides 45 fully-funded studentships from the ESRC available in the South West across twelve different pathways. I was awarded the funding for a MSc + PhD position at the University of Plymouth under the supervision of [Prof. Andy Wills](#). The project was designed to incorporate a combination of advanced computational methods, rigorous experimental research and open science.

2019 - 2020: M.Sc. Advanced Psychology. School of Psychology, Plymouth University.

The Master's course provided extensive training in advanced methods used in the quantitative methods in Psychology. My thesis was investigating human judgements of ambiguous action influenced by person perception, which was supervised by [Prof. Patric Bach](#).

2017 - 2019: B.Sc. Psychology, Plymouth University *During my Bachelor's, I worked as an apprentice with [Dr. Marina Wimmer](#) and [Prof. Andy Wills](#), who supervised my final year project. The project investigated the predictions of formal models of an irrational effect through open science methods (reanalysis of data sets from published and peer-reviewed papers). It won the EPS/BSA Prize Award.*

TEACHING AND TRAINING EXPERIENCE

How to write clean R code This session was delivered as part of a monthly training session in R that is organised and maintained by the local R community.

Estimating Sample Size with Bayes Factors This workshop is a user-friendly and hands-on introduction to estimating sample sizes with a combination of Bayes Factor and Monte Carlo methods.

Better Tables and Better Graphs A how-to on correlation matrices, custom table of descriptive statistics, and publication-quality graphs showing both central tendency and variability (or uncertainty) of your data.

Introduction to Quantitative Analysis with R This is a half-day session (4 - 5 hours) aimed at first-year Doctoral Clinical Psychology students.

Intermediate Guide to Research Methods in R This is an intermediate half-day session (4 - 5 hours) aimed at second-year Doctoral Clinical Psychology students.

SUPERVISORY EXPERIENCE

Undergraduate Dissertations I supervised multiple final-year undergraduate research projects to date.

Peer-to-peer training and support Assist, train and support junior lab members, as well as PhD students, in experimental design, theory, and coding both in R and python.